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PREVIEW

LATTICE CONSTANT, THERMAL EXPANSION, AND ISOTHERMAL COMPRESSIBILITY MEASUREMENTS ON ARGON SINGLE CRYSTALS

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PREVIEW

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I HEREBY RECOMMEND THAT THE THESIS PREPARED UNDER MY SUPERVISION BY **Otis Granville Peterson**

ENTITLED **LATTICE CONSTANT, THERMAL EXPANSION, AND ISOTHERMAL COMPRESSIBILITY MEASUREMENTS ON ARGON SINGLE CRYSTALS**

BE ACCEPTED IN PARTIAL FULFILLMENT OF THE REQUIREMENTS FOR THE DEGREE OF **Doctor of Philosophy in Physics**

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I. INTRODUCTION

The noble gas solids are in many ways ideal materials for the verification of the theory of crystal lattices. They are composed of spherically symmetric atoms that crystallize into a simple face-centered cubic lattice. The interatomic potential can be represented by an essentially two-body force of short range. These characteristics have encouraged the calculation of the thermodynamic properties of the noble gas solids from several models. The theoretical work has in turn focused interest on the experimental determination of these properties.

Among the noble gas solids the most complete information exists for solid argon. Several authors have written excellent review articles on the properties of solid argon and the other noble gases. The earliest of these was co-authored by Dobbs and Jones[1] who compiled the experimental knowledge available at that time (1957) and correlated it with the theory of lattice dynamics. Hollis Hallett[2] in 1961 compiled and critically discussed the experimental properties of all the noble gases for both the liquid and solid states and made comparisons to the law of corresponding states. Very recently Pollack[3] has contributed a very thorough, non-critical compilation of available theoretical and experimental information concerning the noble gases.

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At about the same time Boato[4] produced a less inclusive but more critical survey of the literature.

X-ray measurements of the lattice constant of solid argon were made as early as 1924 by Simon and von Simson[5] and in 1925 by de Smedt and Keesom.[6] These measurements, however, were only at single temperatures. The first comprehensive lattice constant determinations as a function of temperature were performed by Dobbs, et al.[7] in 1956. Their x-ray data were taken in the temperature range from 20°K to 60°K, using micro-crystalline specimens and a powder camera. They obtained accuracy of one part in 1600, but were forced to stop taking measurements at 60°K because of crystal grain growth.

Powder diffractometers have been used more recently by Bolz and Mauer[8] and also by Barrett and Meyer.[9] Barrett and Meyer were able to extend their measurements to 64°K by cold working their specimens mechanically, then annealing them to produce suitable polycrystalline x-ray samples. They also obtained a single crystal measurement at 81°K with the same diffractometers, but with reduced accuracy. Their low temperature measurements were made with estimated errors not greater than $\pm$ 0.0005 Å.

The present work was aimed toward the improvement of this relatively crude experimental knowledge. The lattice constant of an argon crystal has been measured by a precise,

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single crystal, back reflection x-ray technique. These measurements have been carried out in a temperature range from 2.3°K to 83.2°K, determining the 0°K lattice constant, the total thermal expansivity and coefficient of thermal expansion. The lattice constant was determined with an uncertainty of less than 15 ppm and the volume expansivity with an uncertainty of three to six ppm. These figures represent an improvement in the accuracy of the expansivity over published values of a factor of 30 or better.

Also, a measurement of the compressibility of solid argon has been made at 4.2°K with 1.3% precision. This was the first isothermal compressibility measurement ever reported on a noble gas solid at pressures below a few hundred atmospheres and the first for argon below 65°K. This was important because of the very pronounced pressure and temperature dependence of the compressibility.

A cryostat suitable for performing these measurements was designed. It was equipped with a large x-ray window permitting Laue photographs to be taken. The orientation of lattice planes was determined which would produce suitable x-ray diffraction peaks. The temperature of the crystal could be adjusted precisely and kept constant for long periods. The specimen chamber was structurally designed to hold 20 atmospheres of pressure. This permitted compressibility measurements to be performed for volume compressions up to 0.09%.

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Investigations were also undertaken to determine the effects of different growth rates and methods upon crystal perfection and microstructure. The results of this work were crystals of much higher quality than had been previously obtained.[3] The final best crystal was a single 3 mm diameter grain over a distance of 7 mm and had an approximate minimum dislocation density of $10^6/cm^2$.

Finally, many of the thermodynamic properties of solid argon were compiled, using the values of the expansivity and density obtained by the present work. These could be determined with much greater accuracy than was previously possible. This greater accuracy permitted some inferences to be made of the equilibrium vacancy content of the material.

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II. EXPERIMENTAL

The experimental method employed was the measurement of the single crystal lattice parameter as a function of temperature and pressure. Previous experimenters using polycrystalline specimens could not cover the entire temperature range of interest.[3] The present measurements, however, were made over the temperature range 2.3°K to 83.2°K, only 0.6°K under the triple point of the crystal. The oscillating, high angle, back-reflection camera used for the x-ray measurements was the one designed and used by Batchelder and Simmons.[10]

A. Cryostat

The liquid helium cryostat, Fig. 1, was designed with a rigid support for the tail of the liquid helium reservoir. This rigidity was achieved by the use of three concentric cylinders constructed alternately of plastic and aluminium. The plastic members provided thermal insulation as well as physical support. To insure a sealed vacuum for the cryostat, the plastic to aluminium joints were bonded together with an epoxy resin. The Linde Corporation constructed the cryostat to the University of Illinois specifications and insulated it with super insulation. The initial design was worked out by F. E. L. Witt of the University of Illinois in conjunction with the Linde Corporation. The thermal loss through the completed tail assembly, as described below, was low enough

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to permit a hold time of 35 hours for a full load of 4.6 liters of liquid helium.

The specimen chamber, Fig. 2, was designed with a large x-ray window, 0.92 steradians, that allowed the crystal orientation to be determined and the desired reflection located. This window was constructed of beryllium to keep the x-ray attenuation at a minimum. For high pressure usage it was a disk 13 mm in diameter and 1.5 mm thick, mounted 1.5 mm from the crystal. It was sealed into the chamber with an epoxy, R8-2038/H8L-3466, manufactured by the Hysol Corporation. For low pressure usage a thinner window, 0.13 mm x 25 mm diameter, was mounted 13 mm from the crystal.

A visual window has been included in the specimen chamber to permit crystal observation and x-ray camera alignment. It was constructed of Pyrex or Mylar, depending on the desired operating pressure. The glass-to-metal joint was sealed with a covering of Mylar glued to both materials. When not in use this window was covered by an aluminum shutter to prevent radiation heating.

The specimen chamber was constructed in two parts of heat-treated beryllium-copper. A vacuum seal between the two parts of the chamber was produced by an O-ring made of indium wire 1.5 mm in diameter. The ends were smoothly cut at 45° and laid together when the seal was assembled, making a continuous, pressure welded O-ring.

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During compressibility measurements the chamber was pressurized from a cylinder of helium gas by means of a high pressure dome regulator and a simple manifold. The helium gas was purified by a John P. Mickel Associates, Inc. adsorption purifier before being admitted to the specimen chamber. The pressures were measured with a Marsh Instrument Company master test gauge, accurate to $\pm$ 2.5 psi, and were checked with a dead weight tester manufactured by Barnet Instruments, Ltd.

The argon crystal was grown inside a Mylar tube 3 mm in diameter x 24 mm long, constructed of 0.012 mm Mylar sheet rolled on a Teflon form and glued with the previously mentioned Hysol epoxy. This tube was glued at the bottom to the specimen chamber and at the top to a small copper block, using the same epoxy. This copper piece was thermally insulated from the specimen chamber and was equipped with a 200 ohm heater that was operated between 5 ma and 25 ma to raise the temperature of this piece with respect to the specimen chamber. By this means a temperature difference of 1° to 10° could be maintained across the crystal while it was being grown.

At first the specimen chamber was equipped for removing the Mylar tube from the crystal specimen by cutting the tube with a hot wire heated by a capacitor discharge, and then pulling the tube up and off of the crystal. To permit this operation the copper piece containing the gradient heater was

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loosely held in a Teflon channel. Also, the specimen gas inlet capillary was spiraled around this channel. This feature was found to be superfluous and therefore was not used.

B. Temperature Control and Measurement

To obtain the desired measurements it was necessary to control the temperature of the specimen to $\pm$ 0.01°K or better for the duration of the x-ray exposures. These exposures varied in length from 10 minutes to 45 minutes as the temperature was increased from 2.3°K to 84°K.

This precise control was achieved over the entire temperature range by two methods of cooling. In the high temperature range the chamber was cooled by conduction through the three Inconel supports, Fig. 3, connecting the specimen chamber to the tail of the liquid helium reservoir. These tubes supplied sufficient thermal conduction to maintain the chamber at approximately 40°K. The temperature was adjusted and controlled in this region by a 300 ohm heater attached to the top half of the specimen chamber. This heater was operated up to 125 ma to change the temperature of the chamber and up to 8 ma to achieve equilibrium temperatures.

The chamber was cooled initially and held at any temperature below 40°K by a heat exchanger attached to the outside of both parts of the chamber. Liquid helium was drawn toward this exchanger at a controlled rate that could be

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adjusted by an external pumping system. The pumping system, shown schematically in Fig. 4, is sufficiently sensitive and stable so that a setting on the needle valve would hold a particular temperature to the required precision with only occasional adjustments.

The approximate flow rates needed to maintain different temperatures are shown in Table I. A discontinuity occurs in the flow rate near the helium reservoir temperature. A stable temperature could not be maintained within 2° of the reservoir temperature by this cooling method. In this range the temperature oscillated over several degrees. The flow rate needed to hold the specimen chamber at 4.2°K was about twice that necessary to obtain 6°.

Temperatures below 4.2°K were obtained by reducing the pressure over the liquid helium reservoir and drawing the helium through the heat exchanger at a sufficient rate to maintain the specimen chamber at the reservoir temperature. This procedure could be employed since the pressure drop across the heat exchanger and the needle valve was normally less than 2 cm of mercury. With the present equipment, temperatures as low as 2.3°K could be achieved with relative ease.

In addition to the liquid helium heat exchanger discussed, the specimen chamber was equipped with another heat exchanger connected to the lower liquid nitrogen reservoir. This heat

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exchanger was attached only to the top part of the specimen chamber and was used during the argon crystal growth process. The operation of this cooling system was analogous to that of the liquid helium system, except that the higher latent heat of vaporization of nitrogen caused this system to be less stable. Frozen nitrogen in this heat exchanger did not interfere with operation of the other exchanger.

Resistance thermometers were used to determine the temperature of the specimen chamber throughout the entire range. This temperature was continuously recorded by plotting the voltage across an appropriate current-stabilized thermometer on a recorder. To obtain the desired precision a constant voltage was subtracted from the thermometer output by a potentiometer and the difference amplified and recorded. By this means the temperature was measured with a precision of 0.001°K, and time averages of the temperature were obtained by graphical integration.

A platinium resistance thermometer calibrated by the National Bureau of Standards to the NBS 1955 scale was used to determine the temperatures between 15°K and 90°K, and a gas-vapor pressure thermometer was used to determine them below that point. A doped, encapsulated germanium resistance thermometer was calibrated from both of the primary thermometers and then used as the working thermometer in the range from 2.3°K to 30°K. A small pressure, 0.1 torr, of helium gas was

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kept in the specimen chamber during measurements to minimize thermal gradients within the chamber.

The output of the DC amplifier in the temperature recording circuit was connected to a simple 0.4w DC amplifier, Fig. 5, that could be used to power the main specimen chamber heater. This simple setup could be used to control the temperature to $\pm$ 0.1°K with a constant setting on the coolant flow system. This was used to control the temperature while the crystals were being grown and annealed, a process that sometimes required several days of well-controlled temperatures several degrees different from the liquid coolant temperatures.

C. Specimen Preparation

Single crystals were grown using a modified Bridgman technique from Research Grade argon obtained from Matheson Company. This gas was of nominal 99.998% purity and had a stated impurity content (in ppm) of $CH_4 < 1$, $CO_2 < 1$, $H_2 < 5$, $N_2 < 5$, and $O_2 < 5$.

1. Growth Parameters

Table II summarizes the results of ten successive trials at crystal growth under different conditions. Several of the crystals were deposited from the vapor while the majority were directionally solidified from the melt. Various growth rates and annealing times were employed to determine the effect of each of these. In each case the crystal

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was grown in a Mylar tube, along which a small, vertical temperature gradient was maintained. Because of the rather slow growth rates employed, a significant amount of annealing of the lower portion of each crystal occurred during the growth. Column 5 of the table therefore notes an approximate total annealing time for the central portion of each crystal.

After growth, each crystal was annealed at the saturated vapor pressure and was then freed from the container walls by thermal etching to prepare it for analysis by back-reflection Laue photographs. The photographs showed that the crystals solidified from the liquid were significantly better than those deposited from the vapor.

Long term annealing did not seem to have appreciable influence on the quality of the crystals, (Specimens 4 and 5), even though large single crystals could be obtained in this manner. The quality of the subgrain structure of the crystal was much more dependent on the rate at which the crystal was grown. Good crystals could be produced at 1 to 2 mm hr^{-1} provided that nucleation and growth of several grains could be avoided. The method finally adopted (Spec. 7-10) involved the solidifing of a seed crystal, annealing of this seed in contact with the melt, and finally directional solidifing of the crystal, with the solid-liquid interface being nearly plane. To obtain this important plane interface the container walls had to have a thermal conductance equal or less than that of

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the column of argon. This was accomplished by the use of the thin-walled Mylar tubes.

2. Crystal Analysis by Laue Photographs

Laue photographs were taken of each crystal at several locations on the crystal. This was done to determine grain size and to select the portion of the crystal most suitable for x-ray back-reflection measurements (minimum mosaic structure). Approximately 100 Laue photographs were taken of the ten crystals at liquid helium temperature. (It was found that back-reflection Laue photographs could not be obtained above 55°K.)

Fig. 6 shows a comparison of Laue patterns from Specimen 5 (vapor-deposited) and from Specimen 8 (melt-grown). In the case of the vapor-deposited crystal, the 0.1 mm diameter collimator (having 0.4° angular divergence and irradiating about 2 mm^2 of crystal surface) revealed considerable sub-structure within each grain, as well as the presence of several grains, Fig. 6A. On the other hand, a typical solidified crystal, Fig. 6B shows well-formed Laue spots with relatively minor substructure.

As noted in Table II, growth from the vapor may produce a crystal having a very slightly greater (531) lattice parameter than that of a melt-grown crystal, although the small number of crystals examined does not warrant a general conclusion.

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